Andrew Olvera

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RESEARCH INTERESTS

Agile, high-speed robotics in complex and GNSS-denied environments. Integrating robot learning, optimal control, computer vision, and learned world models for robust perception, planning, and coordinated autonomy across heterogeneous robot swarms.

EDUCATION

York College of Pennsylvania Aug 2023 – May 2026
B.S. in Computer Science, York, PA

- Coursework: Data Structures, Algorithms, Linear Algebra, Calculus, Probability & Statistics, Computer Vision, Computer Graphics

Cabrini University Aug 2021 – May 2023
Undergraduate coursework, Radnor, PA; transferred to York College

RESEARCH EXPERIENCE

Starscream: Supervised Pretraining for Route-Conditioned Agile Flight 2026 – Present
Independent research

- Pretrained a 5.84M-parameter causal transformer policy using MPCC expert labels, DAgger, DART, and an auxiliary task-delta prediction head.
- Conditioned the policy on recent state and control history, six ordered gates, privileged vehicle settings, and a test-time speed command.
- In a three-seed eval100 comparison, achieved 66.7% clean-and-timely success with the transformer, versus 52.5% for LSTM and 50.1% for history MLP; inputs and training budgets differed.
- Implemented RL fine-tuning of pretrained flight policies; evaluating benefits of pretraining against matched baselines.

Tells: Opponent Beliefs from Play History in a MicroRTS World Action Model 2026 – Present
Independent research

- Built action-conditioned latent world models and state/action tokenizers for MicroRTS, with complete-state and fog-visible representations.
- Distilled a 16-code opponent belief from fog-of-war history using privileged training targets; improved return prediction, while downstream policy benefits remained inconclusive.
- Trained PMPO actor-critic policies with eight-step imagined rollouts in a frozen learned world, alongside real-engine data collection; compared belief-conditioned and separately trained no-belief policies.

Reward Design in Small-Model RLVR 2025 – 2026
Independent research

- Studied RLVR on Qwen3 0.6B (GSM8K), comparing GRPO and DAPO across group sizes, then binary, structured, and rank-based rewards on a fixed baseline.
- Exploratory single-seed runs reached roughly 69–72% accuracy from a 68% SFT baseline; tracked reward variation as binary rewards saturated.
- Analyzed token-level advantages, dynamic clipping, and reward structure to study when the optimizer retains a useful learning signal.

MANUSCRIPTS IN PREPARATION

A. Olvera. *Starscream: Supervised Pretraining for Route-Conditioned Agile Flight.* In preparation, 2026.

A. Olvera. *Tells: Opponent Beliefs from Play History in a MicroRTS World Action Model.* In preparation, 2026.

A. Olvera. *Reward Design in Small-Model RLVR.* In preparation, 2026.

HONORS

- Dean's List, York College of Pennsylvania · Fall 2025, Spring 2026

INDUSTRY EXPERIENCE

Sapio Sciences May 2025 – Present

Application Engineer Intern, York, PA

- Develop and deliver customer-specific software solutions, combining hands-on engineering with customer-facing implementation.
- Collaborate across teams to translate customer needs into working solutions and coordinate delivery.

Dataforma May 2025 – Jan 2026

Software Engineer Intern, Web Development, York, PA

- Developed full-stack web applications with Vue.js and Kotlin, including refactoring legacy JSP modules into a separate frontend and backend.
- Worked across development and QA teams to implement features, debug defects, execute test plans, and validate deployments.

TECHNICAL SKILLS

ML & RL: PyTorch, Hugging Face Transformers, TRL, vLLM; imitation learning (DAgger, DART), policy gradients (PPO, GRPO, DAPO), RLVR, RLHF/DPO, model-based RL, self-supervised learning, computer vision

Control & Simulation: Control theory, MPC/MPCC (acados), ROS, Flightmare, Isaac Sim, CARLA, Basilisk, MATLAB/Simulink

Languages: Python, C++, C, Java, Kotlin, JavaScript, C#, SQL

Tools: Linux, Git, Docker, Bash, HDF5, WebDataset